

Three-Round Adaptive Diagnosis in Crossed Cubes *

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Abstract

In this paper, we propose a scheme to solve the problem of adaptive fault diagnosis in the n -dimensional crossed cubes in at most three test rounds, provided that the number of faulty vertices is at most n for $n \geq 10$. First, each vertex tests one specific neighbour and is tested by another specific neighbour to provide a basic syndrome in two rounds in our scheme. Then, we assign other necessary tests to diagnose the vertices that cannot be identified according to the previous syndrome in one more round. The scheme is optimal for at most three rounds since the adaptive diagnosis needs at least three rounds to complete.

1 Introduction

Interconnection networks plays a major role in parallel computing systems. Since the increasing size of the multiprocessor system and the longer operation time in the system, the components of the multiprocessor system are easier failed than before. The concept of system-level fault-diagnosis has first been proposed by Preparata, Metze, and Chien [6]. Each processor performs tests of their neighbors in the system and output their test results. By analyzing the complete collection of the test results, the observer identifies which is *faulty* and which is *fault-free*. A characterization of t -diagnosable systems is given by Hakimi and Amin [4].

Nakajima first proposed an adaptive diagnosis model to improve the efficiency of the diagnosis process in [5] which test assignments are dynamically determined after checking the results of previous test results. Fujita and Araki proposed three

round adaptive diagnosis in n -dimensional hypercube [3].

An n -dimensional crossed cube, CQ_n [2], is an important variation of hypercube Q_n and preserves many of its desirable properties. CQ_n has 2^n vertices and $n(2^{n-1})$ links, same as hypercube Q_n . However, the diameter, wide diameter, and faulty diameter in crossed cubes are about half of those in comparable hypercubes.

The paper is organized as follows. The preliminary knowledge and fundamental definitions for the crossed cube, PMC model, t -diagnosable in the next section. Then, we describe an interesting scheme to transform a CQ_n into a hypercube-based hierarchical network in Section 3. A useful technique for detecting the status of each vertex (fault-free, faulty, or suspicious) from the basic syndrome is introduced in Section 4. In Section 5 we introduce our diagnosis scheme and some important concepts for assigning the furthermore necessary tests to identify all faults from the suspicious vertices. Conclusions are given in the final section.

2 Preliminary

For the graph definition and notation we follow [7]. $G=(V, E)$ is a graph if V is a finite set and E is a subset of $\{(u, v) | (u, v) \text{ is an unordered pair of } V\}$. A *path* is a sequence of adjacent vertices written as $\langle v_0, v_1, \dots, v_m \rangle$, in which all the vertices $v_0, v_1, \dots, v_m$ are distinct except possibly $v_0 = v_m$. A *cycle* is a path with at least three vertices such that the first vertex is the same as the last one.

To define crossed cube [2], a notion called “pair related” relation is listed as follows.

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Definition 1 Let $R = \{(00, 00), (10, 10), (01, 11), (11, 01)\}$. Two digit binary strings $u = u_1u_0$ and $v = v_1v_0$ are pair related, denoted as $u \sim v$, if and only if $(u, v) \in R$.

The following is the recursive definition of the n -dimensional crossed cube CQ_n .

Definition 2 The crossed cube CQ_1 is a complete graph with two vertices labelled by 0 and 1, respectively. For $n \geq 2$, a n -dimensional crossed cube CQ_n consists of two $(n-1)$ -dimensional sub-crossed cubes, CQ_{n-1}^0 and CQ_{n-1}^1 , and a perfect matching between the vertices of CQ_{n-1}^0 and CQ_{n-1}^1 according to the following rule: Let $V(CQ_{n-1}^0) = \{0u_{n-2}u_{n-3}\dots u_0 : u_i \in (0, 1)\}$ and $V(CQ_{n-1}^1) = \{1v_{n-2}v_{n-3}\dots v_0 : v_i \in (0, 1)\}$. The vertex $u = 0u_{n-2}u_{n-3}\dots u_0 \in V(CQ_{n-1}^0)$ and the vertex $v = 1v_{n-2}v_{n-3}\dots v_0 \in V(CQ_{n-1}^1)$ are adjacent in CQ_n if and only if

1. $u_{n-2} = v_{n-2}$ if n is even, and
2. $(u_{2i+1}u_{2i}, v_{2i+1}v_{2i}) \in R$, for $0 \leq i < \lfloor \frac{n-1}{2} \rfloor$.

For convenience, let $v = v_{n-1}v_{n-2}v_{n-3}\dots v_0$ be the i -dimensional neighbor, $0 \leq i \leq n-1$, of $u = u_{n-1}u_{n-2}u_{n-3}\dots u_0$ if the following conditions are satisfied: (1) $u_j = v_j$ for $i+1 \leq j \leq n-1$, (2) $u_i \neq v_i$, and (3) $(u_{2j+1}u_{2j}, v_{2j+1}v_{2j}) \in R$ and $u_{i-1} = v_{i-1}$ if i is odd, for $0 \leq j < \lfloor \frac{i-1}{2} \rfloor$.

For $n \geq 3$ and u is a vertex in CQ_n , $V_e(u) = \{v | u_{2i} = v_{2i}, \text{ for } i=0, 1, \dots, \lfloor \frac{n}{2} \rfloor\}$. Let $G[V_e(u)]$ be the subgraph of the CQ_n induced on $V_e(u)$ and let $\text{dif}(u, v)$ be the set of the positions of the different bits between u and v . For an instance, let u be 10001 and v be 00111, $\text{dif}(u, v)$ is $\{1, 2, 4\}$.

2.1 Diagnosis Model

Let u and v be a pair of adjacent vertices of G . The outcome of the test conducted by u and v is denoted by $r(u, v)$. The test result $r(u, v) = 0$ (resp. 1) if u evaluates v as *fault-free* (resp. *faulty*). A test with outcome 0 (resp. 1) is called 0-arrow (resp. 1-arrow).

A test assignment for a system $G = (V, E)$ is a collection of tests (u, v) for some adjacent vertices. The collection of all test results for a test assignment is called *syndrome*. A system is t -diagnosable if there exists a test assignment such that all faulty vertices are identified, provided that the number of faulty vertices does not exceed t [6]. In non-adaptive diagnosis, all test assignment of a system must be scheduled in advance. In

adaptive diagnosis, first proposed by Nakajima [5], the next tests are determined by the previous test results.

3 Architectural relation between crossed cubes and hypercubes

The hypercube is one of the most popular interconnection networks because many parallel algorithms have been developed using this structure. It is well known that an n -dimensional crossed cube CQ_n is an important variation of hypercube Q_n . In this section, we furthermore show that a crossed cube can be transformed into a two-level hypercube-based hierarchical network.

Lemma 1 Let $G(V, E)$ be an n -regular graph with $|V| = 2^n$ and for each vertex, all edges incident to it are labeled with n distinct dimensions, $\{0, 1, \dots, n-1\}$. Then G is isomorphic to a Q_n if for any two neighbours connected by an i -dimensional edge, their neighbours corresponding to two respective j -dimensional edges are also connected by an i -dimensional edge.

Lemma 2 For each vertex u of a CQ_n , $G[V_e(u)]$ is isomorphic to a $Q_{\lfloor \frac{n}{2} \rfloor}$.

Lemma 3 For an integer $n \geq 3$ and two vertices u and v of a CQ_n , $V_e(u) \cap V_e(v) = \emptyset$ if $u_{2i} \neq v_{2i}$ for some integer $0 \leq i \leq \lfloor \frac{n}{2} \rfloor$.

Proof: Let any vertex a of the $V_e(u)$. $a_{2i} = u_{2i} \neq v_{2i}$. Since $a \notin V_e(v)$, $V_e(u) \cap V_e(v) = \emptyset$.

To simplify the explanation, if no ambiguity arises, for an integer $n \geq 3$, $m = \lfloor \frac{n}{2} \rfloor$ and a vertex u of a CQ_n , consider the $G[V_e(u)]$ as a super vertex, denoted as S_u .

Lemma 4 For an integer $n \geq 3$, an even integer i and two vertices u and v of a CQ_n with $\text{dif}(u, v) = \{i\}$. There is an one-to-one and onto function $f : V_e(u) \rightarrow V_e(v)$ by one vertex of the $V_e(v)$ being an i -neighbor of the $V_e(u)$ in the S_u and the S_v .

We say that S_u and S_v is adjacent if there is an one-to-one and onto function.

Theorem 1 The structure of CQ_n consists of two levels of hierarchies. For an integer $m = \lfloor \frac{n}{2} \rfloor$, there is 2^{n-m} classes connected in a hypercube fashion in the upper level of hierarchy, and each class is formed as a Q_m of 2^m vertices in the lower level.

Proof: By lemma 2 and 3, each class is formed as a Q_m . By lemma 4, all classes are connected in a hypercube fashion.

Since a CQ_n can be transformed into a two-level hypercube-based hierarchical network, we introduce a lemma associated with the Q_n below which will be used in the following sections.

Lemma 5 *For any subset $U \subseteq V(Q_n)$ of cardinality n , there is a set of vertices $W (\subset V(Q_n) - U)$ of cardinality n such that there is a matching of size n between U and W in Q_n .*

4 Cycle Partition

In section 3, we show how to transform a CQ_n into 2^{n-m} m -cubes. Note, each m -cube contains a Hamiltonian cycle. Let C be the Hamiltonian cycle. Then we can get all the basic test results in C in two rounds. If we number the vertices in C , then the vertices of even number can test the vertices of odd number in the first round and the vertices of odd number can test the vertices of even number in the second round. In this section, we introduce an algorithm to partition a cycle into paths according to the test results obtained in the first two rounds in C .

Algorithm 1 [6] CYCLE-PARTITION

Step 1: Choose a 0-arrow r_0 followed by a 1-arrow.

Step 2: Let r be the arrow following r_0 . If r is 0-arrow, then set $r_0 \leftarrow r$ and proceed step 2; otherwise, go to step 3.

Step 3: Mark with an X the arrow following 1-arrow r . If it was not already marked, set r_0 the X-marked arrow and go to step 2; otherwise, the algorithm is terminated.

Now we can partition C into paths by removing the test result marked by X.

For example, as shown in Fig.1, a cycle with sixteen vertices can be partitioned into three paths.

Lemma 6 [6] *Each path contains at least one faulty vertex.*

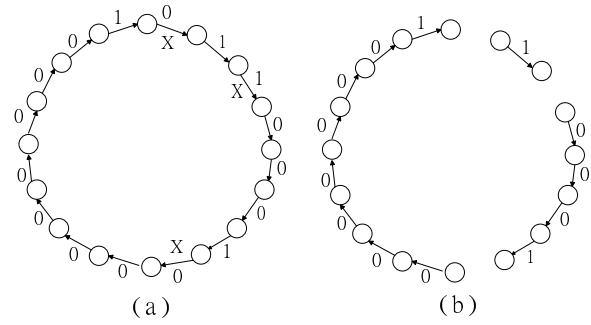


Figure 1: (a) Before CYCLE-PARTITION (b) After CYCLE-PARTITION

5 Adaptive diagnosis scheme

In this section, we show that how to perform the diagnosis of the crossed cube in at most three rounds if $m \geq 5$ where $m = \lfloor \frac{n}{2} \rfloor$.

5.1 First Partition

At first, transforming the CQ_n into 2^{n-m} m -cubes, $H_1, H_2, \dots, H_{2^{n-m}}$, and then executing the first two round tests in each m -cube's Hamiltonian cycle. Since $|V(Q_m)| \geq n + 1$, any m -cube with no 1-arrows must be fault-free. Any m -cube with at least one 1-arrow must contain at least one faulty vertex. Let x : the number of m -cubes which contain at least one 1-arrow and let $S_1 = \{H_1, H_2, \dots, H_x\}$ be the set of these m -cubes. In this paper, we suppose that there is an observer who can collect all the number of the paths in S_1 after CYCLE-PARTITION. Let s_i : the number of paths in H_i and let $s = \sum_{i=1}^x s_i$. Next two subsections discuss how to identify faults according to the relation between x and $n - m$.

5.2 $x \leq n - m$

Note, a CQ_n can be treated as a Q_{n-m} of 2^{n-m} super vertices and S_1 can also be viewed as a set with x super vertices. Since $x \leq n - m$, by lemma 1, there is a set $W \subset V(Q_{n-m}) - S_1$ such that there is a perfect matching of size $|S_1|$ between $|S_1|$ and W . Since each m -cube in W has been known as fault-free, each m -cube in S_1 can be tested by a fault-free m -cube, and the whole diagnosis will be done in three rounds.

5.3 $x > n - m$

Each m -cube in S_1 must contain at least one and at most m faulty vertices (In fact, if it contains $m + 1$ fault vertices, the total number of the fault vertices in the CQ_n will be at least $m + 1 + (x - 1) \geq n + 1$, a contradiction). According to lemma 6 and CYCLE-PARTITION, we show an algorithm to distinguish the vertices of the m -cubes in S_1 .

Algorithm 2 [1] PATH-COLORING

let $q = n - s + 1$. We give a color to each vertex in each path as follows.

1. If path has more than $q + 1$ vertices and has at least one 0-arrow, then we color the head of the path with black and color $q - 1$ vertices from the tail with grey, and all vertices that are not colored yet are colored white.
2. If path has at most $q + 1$ vertices, then we color all vertices in path with grey.

The white vertices must be fault-free, the black vertices must be faulty, and the grey vertices needs to be tested for identifying their status. Fig.2 provides the results of performing PATH-COLORING on the three paths of Fig.1(b).

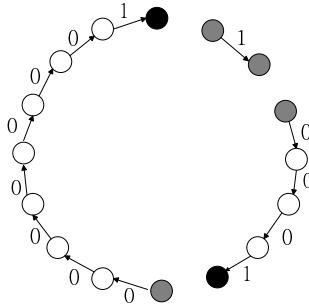


Figure 2: An example of PATH-COLORING

Consider a Hamiltonian cycle constructed in the first two rounds, a (consecutive) part of the Hamiltonian cycle in a m -cube is called a *block*. A block is said to be “dirty” if it contains vertices to be colored with black or grey by PATH-COLORING and said to be “clean” if all the vertices in the block to be colored with white. Note that any vertex of the clean block is fault-free. Now we know that there are at least s faulty vertices in the CQ_n . By lemma 6, each

path contains at most $n - s + 1$ faulty vertices. Now let $l = \lceil \log_2(n - s + 1) \rceil$ and 2^l : the size of the block. Hence we can partition a m -cube into 2^{m-l} l -cubes, $B_1, B_2, \dots, B_{2^{m-l}}$. Note that a path can produce at most two consecutive dirty blocks.

After counting, we have known the number of dirty blocks in each m -cube. Since $x > n - m$, it is possible that not all of the m -cubes in S_1 can be tested by the corresponding clean m -cubes. However, we hope some m -cubes can be diagnosed at first. In general, a m -cube contains more paths has less chances to be diagnosed by itself. So we can let m -cubes which owns more paths have higher priority to be tested by a clean m -cube. This can reduce the complexity of the whole diagnosis scheme. In other words, it means we can first make a maximum matching to let most of m -cubes be diagnosed with their respective clean neighbors in the three rounds. The remaining m -cubes cannot be matched with clean neighbors are discussed in the two cases below.

Case 1: If the number of dirty blocks in a m -cube doesn't exceed $m - l$, similar to Section 5.2, the whole diagnosis will be done in three rounds.

Case 2: If the number of dirty blocks in some m -cube exceeds $m - l$, let S_2 be the set of such m -cubes. If a m -cube has only one or two dirty blocks, it's useful for the following discussion. We know that a m -cube with exactly one vertex has exactly one dirty block or two consecutive dirty blocks. So a m -cube with exactly one faulty vertex can have at least $2^{m-l} - 4$ (there are at most two clean blocks are used to test the dirty block(s) in that m -cube) clean blocks to be used. We can use these m -cubes to test the m -cubes in S_2 . Since the number of m -cubes which contains two or more faulty vertices doesn't exceed $m - 1$ (if exceed $m - 1$, the total number of the fault vertices in CQ_n will be at least $2m + (n - m + 1 - m) > n$, a contradiction), each m -cube in S_2 has at least $n - m - (m - 2) = n - 2m + 2$ adjacent m -cubes which contains at most one faulty vertex. Note that $n - 2m + 2 \geq 2$. Hence we can assign two such m -cubes for each m -cube in S_2 .

Let H_2 and H_3 be the two such m -cubes and H_1 be a m -cube in S_2 (The worst case is that two blocks in H_1 can't be tested by the blocks in H_2 and H_3 , and such a situation occurs only when

the dirty blocks in H_2 and H_3 are adjacent with those in H_1), we can use the clean blocks in H_2 and H_3 to test the dirty blocks in H_1 . If the worst case occurs, let I be the set of those two dirty blocks in H_1 . Since each m -cube contains at most m paths, we next discuss how to identify faults according to the relation between m and s_1 of H_1 in S_2 as the following subcases.

case 2.1 : $s_1 = m$

That is, H_1 has m faults. Since H_1 cannot be matched with a clean m -cube, all adjacent m -cubes of H_1 have faulty vertices. If there is a adjacent m -cube which has two or more faulty vertices, the total number of the faulty vertices in the CQ_n will be $m + (n - m - 1) + 2 > n$, a contradiction. Hence each adjacent m -cube of H_1 has exactly one faulty vertex. Assigning a such m -cube to H_1 and almost all of the grey vertices in H_1 can be tested by the respective neighbors in the assigned m -cube. In the worst case, only one vertex can't be identified if its neighbor in the assigned m -cube is faulty. Since H_1 has m faults, each path of H_1 has exact one fault. That is, a grey vertex must belong to a path with only one edge. Note, such path has one 1-arrow and two end vertices of the 1-arrow are both grey. Clearly, one end vertex of the 1-arrow is faulty and another one is fault-free. Therefore, we only need to check the adjacent vertex of the unidentified vertex in the same 1-arrow. And the whole diagnosis will be done in three rounds.

Case 2.2 : $2 \leq s_1 \leq m - 3$

For all the adjacent m -cubes of H_1 have faulty vertices, $(n - m) + 2 \leq s \leq n$. If $s = m$, it's similar to the case of 2.1. If $(n - m) + 2 \leq s \leq n - 1$, $1 \leq l \leq \lceil \log_2(m - 1) \rceil$ and $m - l > s_1$ for $m \geq 5$. Each path can produce at most two consecutive dirty blocks. But we know that the adjacent blocks of the dirty block in I are not consecutive and each block in the m -cube has $m - l$ adjacent blocks, so each path contains at most one adjacent dirty block of the dirty block in I . Since $m - l > s_1$, it must exist at least one adjacent clean block of the dirty block in I . Hence the dirty blocks in I can be tested by the clean block(s). The whole diagnosis will be done in three rounds.

Case 2.3 : $s_1 = m - 2$

For all the adjacent m -cubes of H_1 have faulty vertices, $s \geq m - 2 + (n - m) = n - 2$. If $s = n - 1$, $l = 1$ and $m - l > s_1$, it's similar to the case 2.2. If $s = n$, it's similar to the case 2.1. If $s = n - 2$, then $l = 2$ and $m - l = s_1$ for $m \geq 5$. The dirty blocks in I can probably have adjacent clean blocks in m -cube, hence we first try to find if there is such clean blocks. If they exist, those dirty blocks can be tested by them and the whole diagnosis will be done in three rounds. If they don't exist, we have to discuss more deeply. Let B_i be such dirty block in I . Since each adjacent block of B_i is dirty, hence B_{i-1} and B_{i+1} must be dirty. Such three consecutive dirty blocks, B_{i-1} , B_i and B_{i+1} , must belong to at least two paths. And because $m - l = s_1$, B_i have exactly $m - l - 2$ adjacent blocks(not B_{i-1} and B_{i+1}) which belong to $s_1 - 2$ paths. Obviously, B_{i-1} , B_i and B_{i+1} must belong to exactly two paths. Now we can list all the case which can produce the situation described above and check if the whole diagnosis will be done in three rounds. By observing all the cases, they all will be done in three rounds. Although there are sixteen cases, we only show some in Fig3 for reducing paper length.

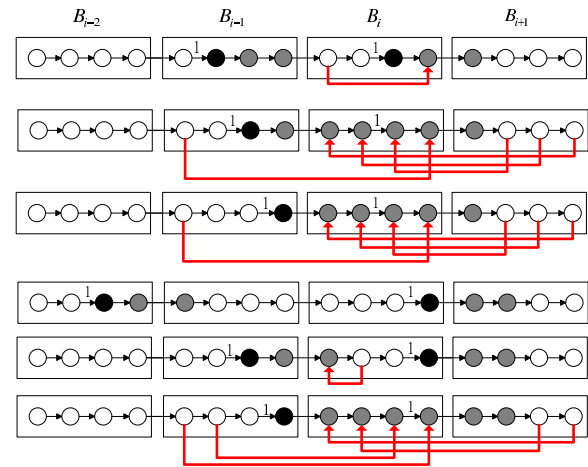


Figure 3: Case 2.3

Case 2.4 : $s_1 = m - 1$

For all the adjacent m -cubes of H_1 have faulty vertices, $s \geq m - 1 + (n - m) = n - 1$. If $s = n$, it's similar to the case 2.1. If $s = n - 1$, $l = 1$, it's similar to case 2.3. By listing all the cases, the whole diagnosis will be done in three rounds. All the seven cases are shown in Fig.4.

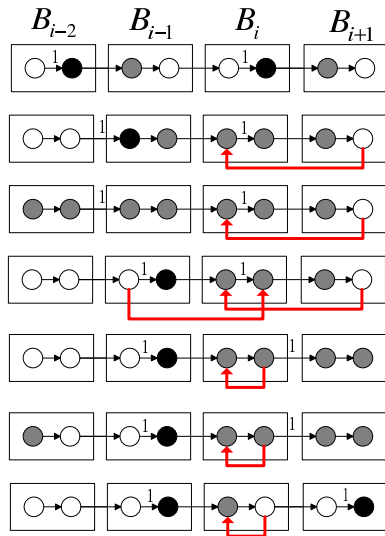


Figure 4: Case 2.4

6 Conclusion

In this paper, we propose a scheme of adaptive diagnosis in the crossed cube with at most n faulty vertices for $n \geq 10$. And the whole diagnosis completes in at most three rounds through the scheme.

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